

Automotive Holographic Head-Up Display

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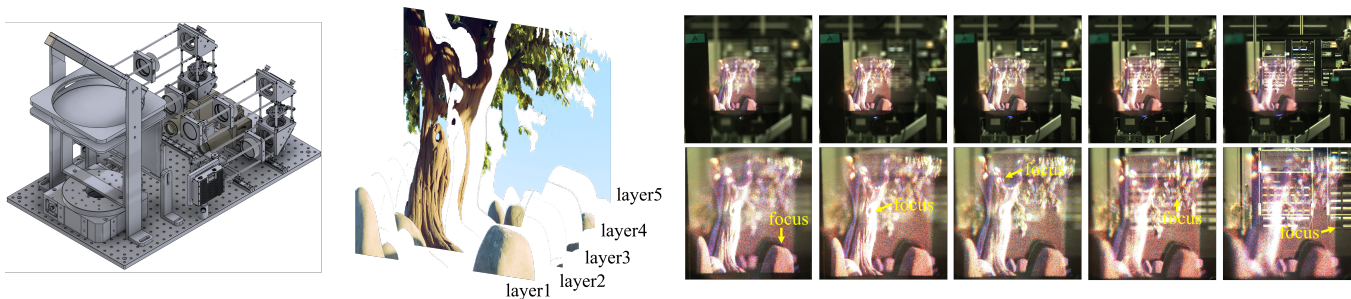


Figure 1: (a) Perspective 3D view of developed automotive holographic head-up display, (b) Depth layer schematics for hologram generation, and (c) Optically reconstructed results of hologram

Abstract

This paper presents an advanced automotive holographic head-up display (HUD) system for providing three-dimensional holograms, extended virtual image distance ($>10\text{m}$), large field of view (>15 degrees), and wide monocular eye-box ($>2\text{cm}$) without distortion and optical aberrations. Our approach ensures high-quality holographic image projection and comfortable viewing, providing a practical solution for next-generation automotive HUDs.

CCS Concepts

• Hardware → Displays and imagers; • Human-centered computing → Displays and imagers.

Keywords

Holographic display, Head-up display, Augmented reality, Automotive

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1 Introduction

In recent years, digital holography and heads-up displays (HUDs) have experienced significant advancements [Christmas and Smeeton 2021]. These technologies are used in various industries, from entertainment to medical imaging, enhancing usability and safety in applications like aviation, automotive, and augmented reality (AR). The integration of digital holography into HUDs promises natural 3D experiences.

Automotive HUDs are categorized into combiner (1st gen.), windshield (2nd gen.), and AR HUDs (3rd gen.). AR HUDs also use the windshield, offering over 15 degrees of field of view (FoV) and virtual image distance (VID) greater than 5 meters, but cannot yet display 3D content. Such a technical development flow has led to the attention now being focused on holographic HUDs, which represent the next generation of AR HUDs, known for their ability to adjust content depth position.

This study aims to describe the technical and novel approaches to enhancing optical specifications for next-generation automotive holographic HUDs.

2 Hardware Overview

As shown in Figure 2, our holographic HUD system consists of two main units: the hologram projection unit (HPU) and the windshield unit (WU). The HPU uses Fourier holography technique to generate approximately 5-inch sized intermediate holograms. The windshield unit delivers these holograms at the pupil, 600mm position from the windshield.

2.1 Optical System Configuration

As shown in Figure 1 (a) and Figure 2, the HPU is composed of 4f optics and a DOE for expanding eye-box. The 4f optics system consists of two lenses with different focal lengths: the first inverse

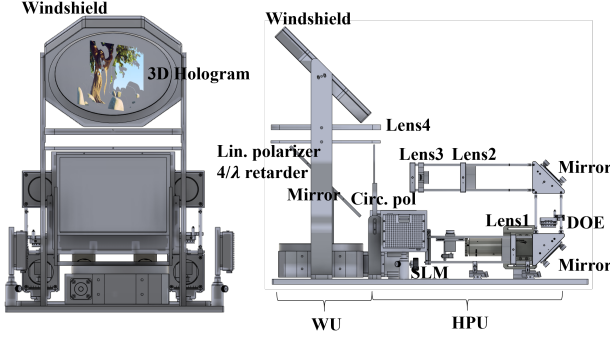


Figure 2: Design of developed automotive holographic head-up display, (a) front view and (b) side view.

Fourier lens (Lens1) has a focal length of 200 mm, while the second Fourier lens (Lens2) has a focal length of 100 mm. This configuration reduces the size and pixel pitch of the spatial light modulator (SLM) by half. Consequently, the diffraction angle increases, resulting in a FoV of approximately 15 degrees. Then the third Fourier lens (Lens3), which has a 500 mm focal length, projects the intermediate hologram to the WU.

The WU consists of folded optics and a field lens. The folded optics use polarization elements to get a sufficient optical path length. The field lens (Lens4) is designed with free-form surfaces to compensate for the curvature and aberrations of the windshield.

3 Optical System Optimization

The field lens (Lens4) in our system is designed with free-form surfaces to correct distortion and optical aberrations. However, it was not enough to remove all distortions and aberrations completely. Additionally, reducing the pixel pitch to widen the FoV resulted in a reduced eye-box size, causing difficulty in viewing the holograms. To address these issues, we developed distortion and aberration removal method, and an eye-box expansion method.

3.1 Distortion and Aberration Removal

In our system, both geometric distortions (misalignment, rotation, scaling, etc.) and optical aberrations (tilt, spherical aberration, coma, astigmatism, etc.) occurred simultaneously, necessitating a process to correct both types of errors. For distortion correction, capturing the projected hologram and calculating a homography transformation matrix to warp the image and correct the distortions. In order to remove optical aberration, optimization was proceed to find out Zernike aberrations coefficients by adjusting errors between the desired hologram and captured hologram. This feedback loop continues until L2 loss is minimized.

$$\text{Minimize } \sum_{i=1}^n \left(I_{\text{captured}}(x_i, y_i) - I_{\text{desired}}(x_i, y_i) \right)^2 \quad (1)$$

where I_{captured} represents the intensity of the captured image, I_{desired} represents the intensity of the desired image, and (x_i, y_i) are the pixel coordinates.

3.2 Eye-Box Expansion

Due to the optical reduction of pixel pitch by half for FoV expansion, the initial configuration resulted in an eye-box of 6.5 mm, which is insufficient for comfortable viewing. To address this, our system placed a DOE at the Fourier plane. Traditional methods tile small eye-boxes together, needs bandwidth limitation or real-time hologram recalculations, which add complexity and computational overhead. Our approach ensures robust eye-box expansion by overlapping hologram replications. The DOE creates replicas that overlap by half horizontally, ensuring coherent behavior as a single entity. For proper functioning, an stochastic gradient descent optimization was applied.

4 Hologram Generation

To generate the holograms, we used 3D models from the Blender demo files (<https://www.blender.org/download/demo-files/>). RGB and depth images were extracted, and the depth information was used to divide the RGB images into five depth layers as shown in Figure 1 (b). Each layer was transformed into a hologram using Fourier transformation and quadratic phase multiplication, with optimizations to correct distortions, aberrations, and eye-box issues. The process can be summarized as follows:

1. Depth Layer Division:

$$I_{\text{layer}_i}(x', y') = \begin{cases} I_{\text{RGB}}(x, y) & \text{if } z(x, y) = z_i \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

where $I_{\text{layer}_i}(x', y')$ is the processed layer image at the scaled coordinates, and the scaling factor s_i is applied which is determined by the depth position z_i . The scaled coordinates are defined as $x' = x \cdot s_i$ and $y' = y \cdot s_i$. $I_{\text{RGB}}(x, y)$ is the original RGB image, and z_i represents the depth layer positions (85 cm ~ 10 m). The depth map $z(x, y)$ is used to segment the RGB image into corresponding depth layers.

2. Hologram Generation:

$$H(x, y) = \sum_{i=1}^n \mathcal{F}\{I_{d_i}(x, y)\} \cdot \exp\left(i\pi \frac{(u^2 + v^2)}{\lambda z_i}\right) \quad (3)$$

where $\mathcal{F}\{\cdot\}$ denotes the Fourier transform, $H(x, y)$ is the final combined hologram, and z_i is the depth position of each layer. Here, λ represents the wavelength of the light source, and u, v are the spatial frequency coordinates in the Fourier domain.

Acknowledgments

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